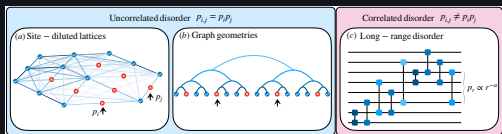


Scalable spin squeezing on disordered quantum networks

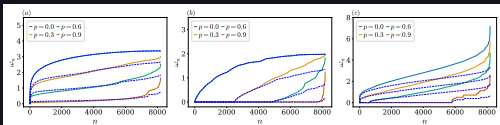
The challenge: Robust spin squeezing on inhomogeneous networks

- Quantum sensors need **entangled states** for precision beyond classical limits
- Realistic platforms have **disorder and imperfections** affecting correlations
- When and how can scalable squeezing emerge in complex, noisy geometries?



Method: Rotor-spin-wave theory on graphs

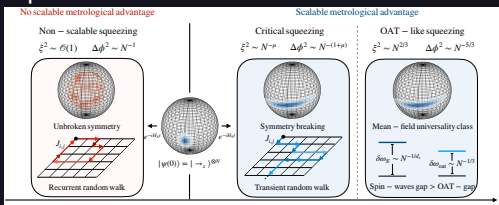
- **Key idea:** Model spin dynamics on general interaction networks via **graph Laplacian spectrum**
- Separate dynamics into:
 - **Zero mode** $\rightarrow$ collective rotor $\rightarrow$ OAT-like squeezing
 - **Spin-wave excitations** $\rightarrow$ governed by Laplacian gaps and spectral dimension



Results: Spectral dimension controls squeezing scaling

$d_s > 3$ leads to OAT-like squeezing
 $2 < d_s < 3$ yields critical squeezing
 $d_s < 2$ no scalable squeezing

- Geometric prerequisite: Existence of a percolating cluster in the network
- Disorder shifts squeezing threshold, but does not affect scaling universality
- Spectral dimension governs dynamic timescales and squeezing minimum scaling exponents



Takeaway: Universality of scalable squeezing on quantum networks

Spin squeezing scalability is *universal* and fully characterized by spectral dimension & graph connectivity.

Disorder-induced critical points result from interplay between percolation and magnetic universality classes.

Path forward: Tunable anisotropy & engineered interactions unlock scalable metrology even in low-filling, disordered systems.